作业 2：下载 Qwen3 0.6B 模型， 本地运行 01_Qwen3.ipynb；

1. 下载 Qwen3 0.6B 模型至本地
2. 修改路径

```
# local_dir = Path(repo_id).parts[-1] # # 本地目录名
local_dir = "../../models/Qwen/Qwen3-0.6B/"
```

```
# 从本地加载权重文件
if CHOOSE_MODEL == "0.6B":
    # 小模型只有一个权重文件
    weights_file = os.path.join(local_dir, "model.safetensors")
    if not os.path.exists(weights_file):
        raise FileNotFoundError(f"本地模型文件不存在: {weights_file}, 请确认模型已正确下载")
    weights_dict = load_file(weights_file)
else:
    # 大模型有多个分片文件，需要合并
    repo_dir = local_dir
    index_path = os.path.join(repo_dir, "model.safetensors.index.json")

    if not os.path.exists(index_path):
        raise FileNotFoundError(f"本地模型索引文件不存在: {index_path}, 请确认模型已正确下载")

    with open(index_path, "r") as f:
        index = json.load(f)

    weights_dict = {}
    for filename in set(index["weight_map"].values()):
        shard_path = os.path.join(repo_dir, filename)
        if not os.path.exists(shard_path):
            raise FileNotFoundError(f"本地模型分片文件不存在: {shard_path}, 请确认模型已完整下载")
        shard = load_file(shard_path)
        weights_dict.update(shard)
```

```
tokenizer_file_path = "../../models/Qwen/Qwen3-0.6B/tokenizer.json"

# if USE_REASONING_MODEL:
#     tokenizer_file_path = f"Qwen3-{CHOOSE_MODEL}/tokenizer.json"
# else:
#     tokenizer_file_path = f"Qwen3-{CHOOSE_MODEL}-Base/tokenizer.json"

# hf_hub_download(
#     repo_id=repo_id,
#     filename="tokenizer.json",
#     local_dir=local_dir,
# )

# 检查本地 tokenizer 文件是否存在
if not os.path.exists(tokenizer_file_path):
    raise FileNotFoundError(f"本地 tokenizer 文件不存在: {tokenizer_file_path}, 请确认模型已正确下载")
```

3. 运行结果

```
D:\Software\Anaconda\anaconda3\envs\py312\python.exe "D:/Software/Pycharm/PyCharm 2023.1.5/plugins/python/helpers/pydev/pydevd.py" --multiprocess --qt-support=auto --client 127.
D:\Software\Pycharm\PyCharm 2023.1.5\plugins\python\helpers\pydev\pydevd_plugins\__init__.py:2: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.i
  __import__('pkg_resources').declare_namespace(__name__)
Connected to pydev debugger (build 231.9414.12)
huggingface_hub version: 0.36.0
tokenizers version: 0.22.1
torch version: 2.6.0
Total number of parameters: 751,632,384

Total number of unique parameters: 596,049,920
float32 (PyTorch default): 5.64 GB
bfloat16: 2.82 GB
<think>
Okay, the user wants a short introduction to large language models. Let me start by recalling what I know. Large language models are AI systems that can understand and generate

I should mention their ability to understand and generate text, not just specific tasks. Maybe include examples like chatbots or language assistants. Also, emphasize their adapt
</think>

Large language models (LLMs) are AI systems designed to understand and generate human language, enabling tasks like text generation, translation, and answering questions. They a
Process finished with exit code 0
```